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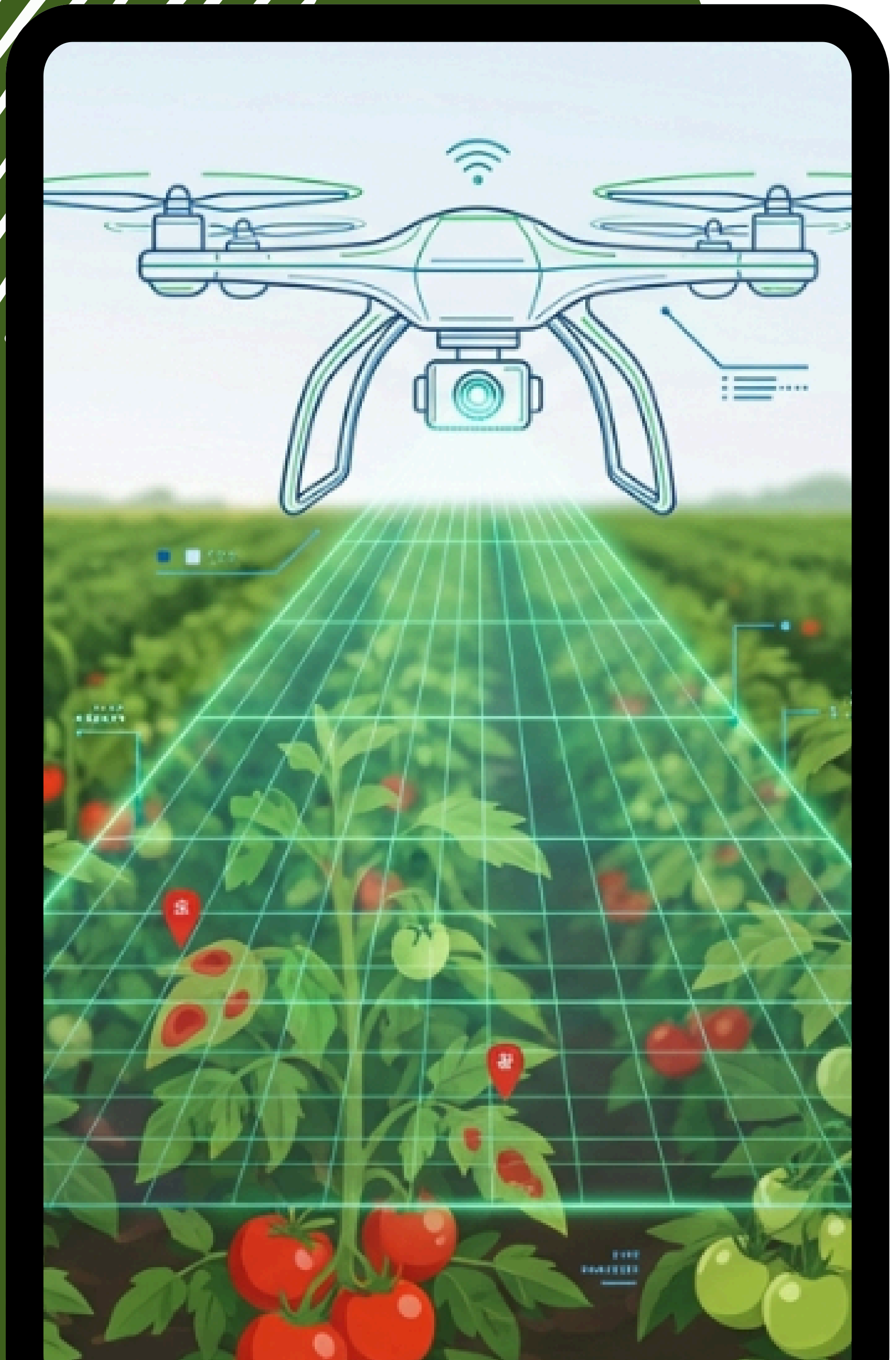


SmartLeafDetection

Hybrid_Deep_Model_for_Tomato_Leaf_Disease_Detection

Project code: 26-1-D-6

Students: Noy Ben Ezra
Ariel Laniado

Advisors: Dr. Dan Lemberg
Mrs. Elena Kramer

Engineering Problem Definition and Project Objectives

- Tomato crops are highly vulnerable to foliar diseases
- Late or inaccurate detection leads to yield loss and increased costs
- Manual inspection is labor-intensive and not scale to large fields

Project objectives:

- Enable fast, large-scale field monitoring
- Detect tomato leaf diseases at early stages
- Produce reliable plant-level diagnostic results

Current Methods and Existing Solutions

The current method

- Requires trained personnel
- relies on visual assessment of plant condition
- Performed periodically across the field

Existing automated approaches

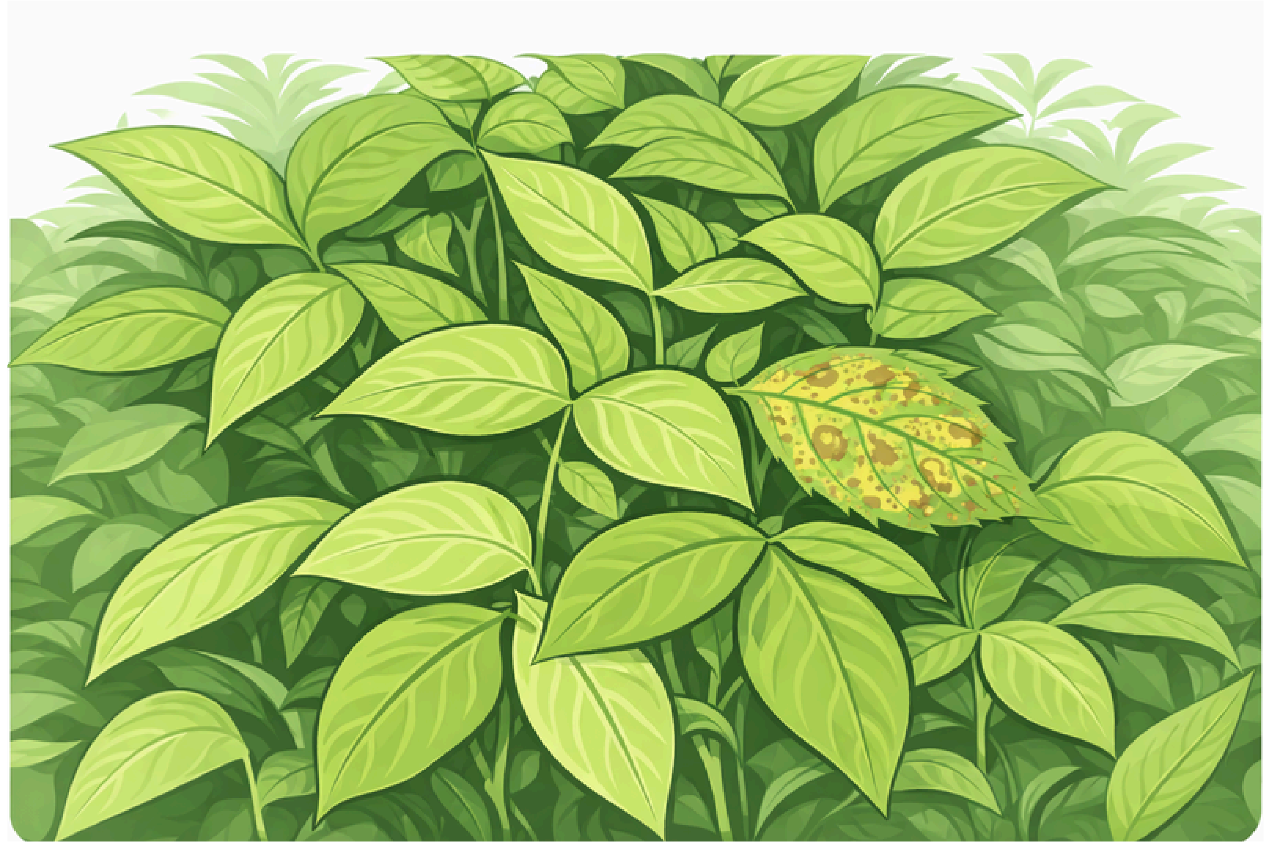
- Image-based leaf disease classification
- Typically rely on single, isolated leaf images
- Developed mainly under controlled conditions

Technical Limitations of Existing Automated Approaches

- Rely on manually captured, close-up leaf images
- Assume clearly visible and isolated leaves
- Depend on human visual judgement for disease detection
- Do not scale to large agricultural fields

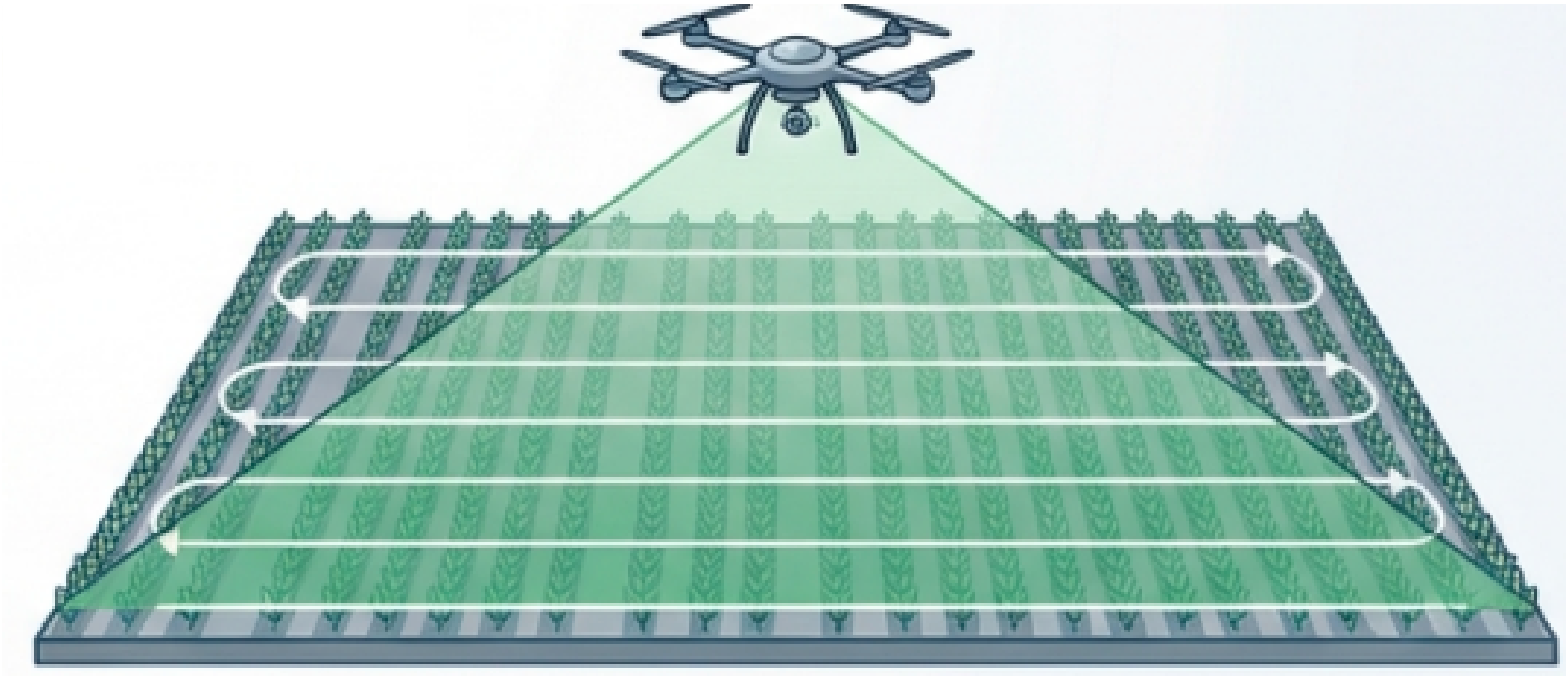
Closeup leaf image

Field conditions



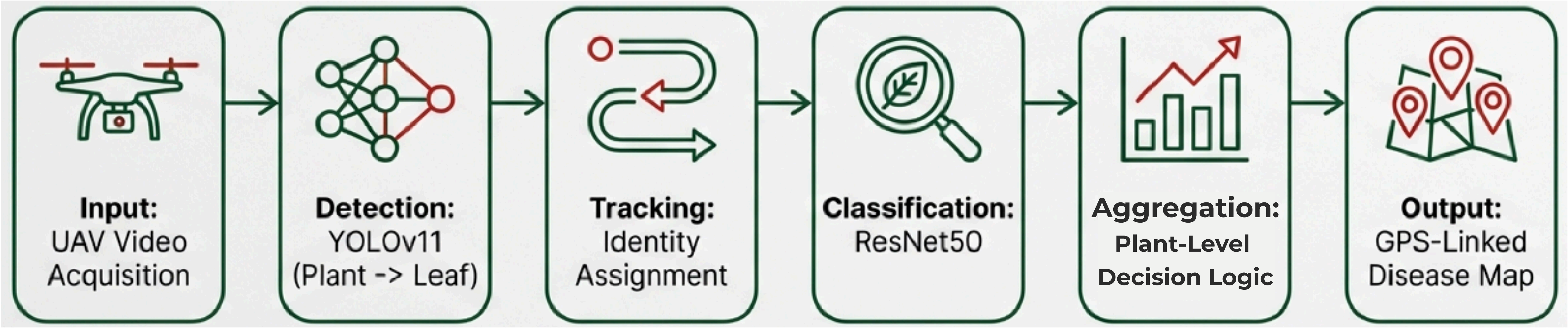
Proposed Solution and Design Considerations

- UAV video acquisition for large scale field monitoring
- Automated detection and analysis of tomato leaves
- Field-wide disease assessment
- Location based diagnostic output



System Processing Pipeline

End-to-end processing from UAV video to field level diagnosis



- UAV video is captured over the tomato field
- Plants and leaves are automatically detected and tracked
- Leaf-level predictions are aggregated into plant-level decisions
- Field-wide disease distribution is generated using GPS data

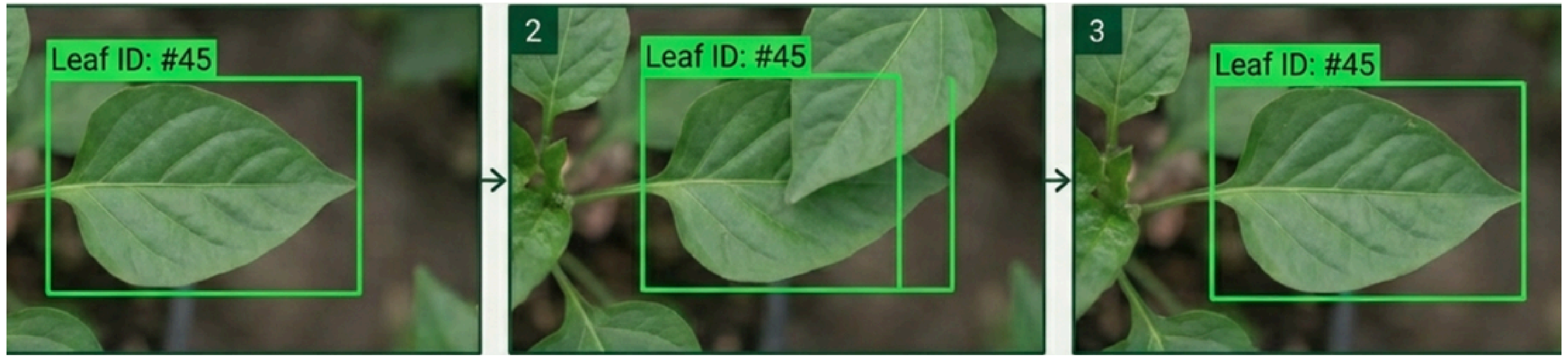
Two-Stage Detection and Multi-Object Tracking

Two-stage Detection

- **Stage A:** Detect tomato plants at field level
- **Stage B:** Detect individual leaves inside each plant region
- Reduces background noise and focuses analysis on relevant regions

Multi-Object Detection and identity

- Plants and leaves are tracked across video frames
- Each detected object is assigned a persistent ID
- Tracking maintains despite motion, partial occlusion and viewpoint changes
- Enables accumulation of evidence over time for reliable diagnosis



Disease Classification and Plant-Level Decision Logic

Disease Classification

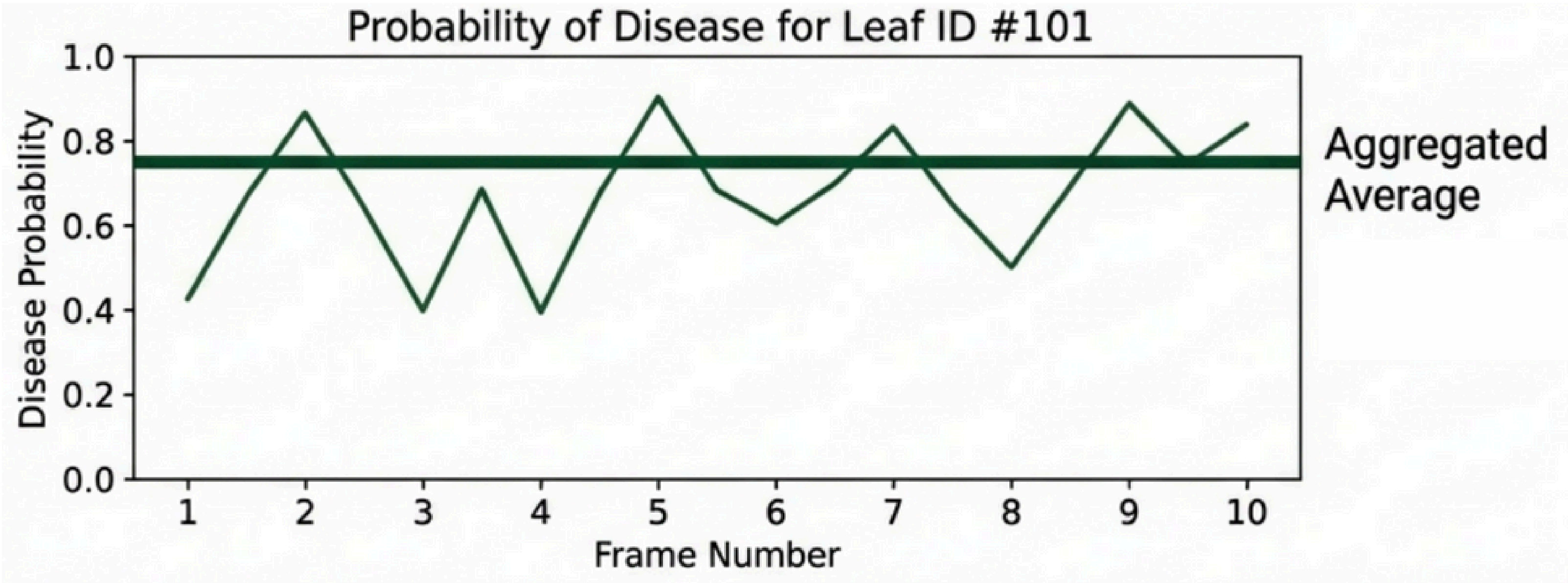
- Each detected leaf ROI is resized and analyzed using a CNN (ResNet50)
- The model outputs disease probabilities for each leaf instance
- Classification is performed independently for every tracked leaf

Plant-Level Decision Logic

- Leaf-level predictions are accumulated across multiple frames
- Final plant diagnosis is derived from aggregated leaf evidence
- Plants may be classified as healthy, diseased or multi-diseased

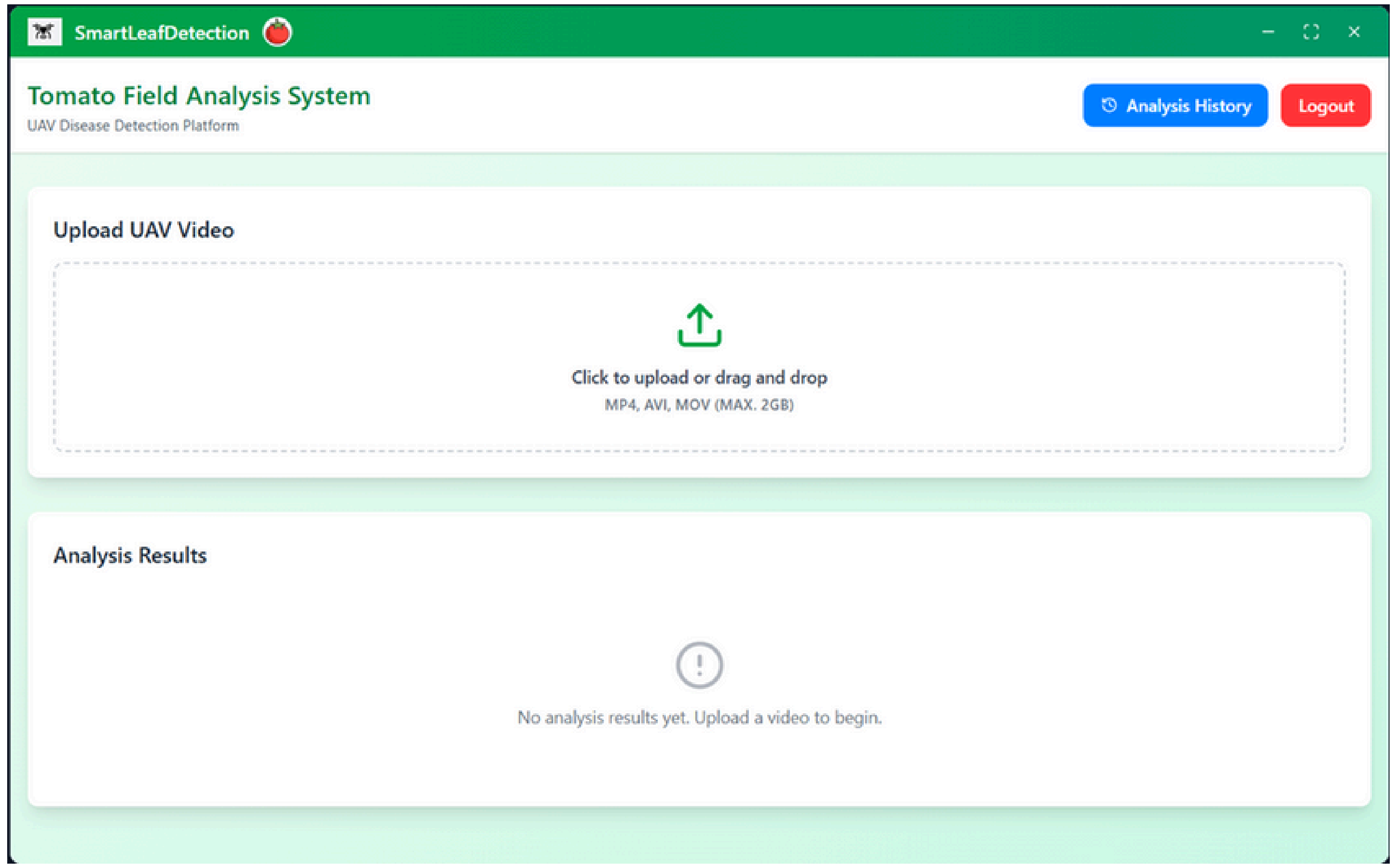
Cropped Leaf (224x224 px)

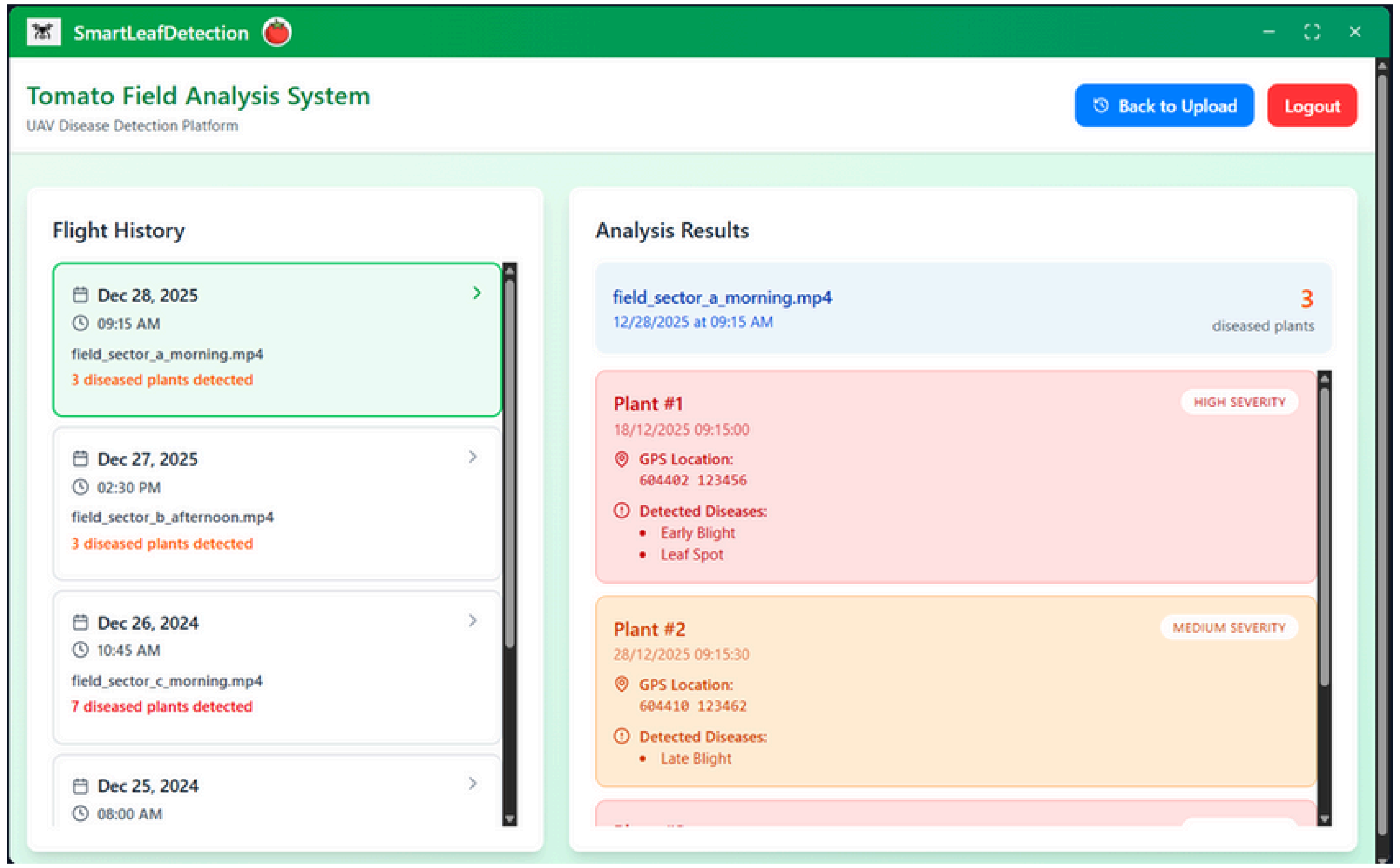
ResNet50
(Transfer Learning)



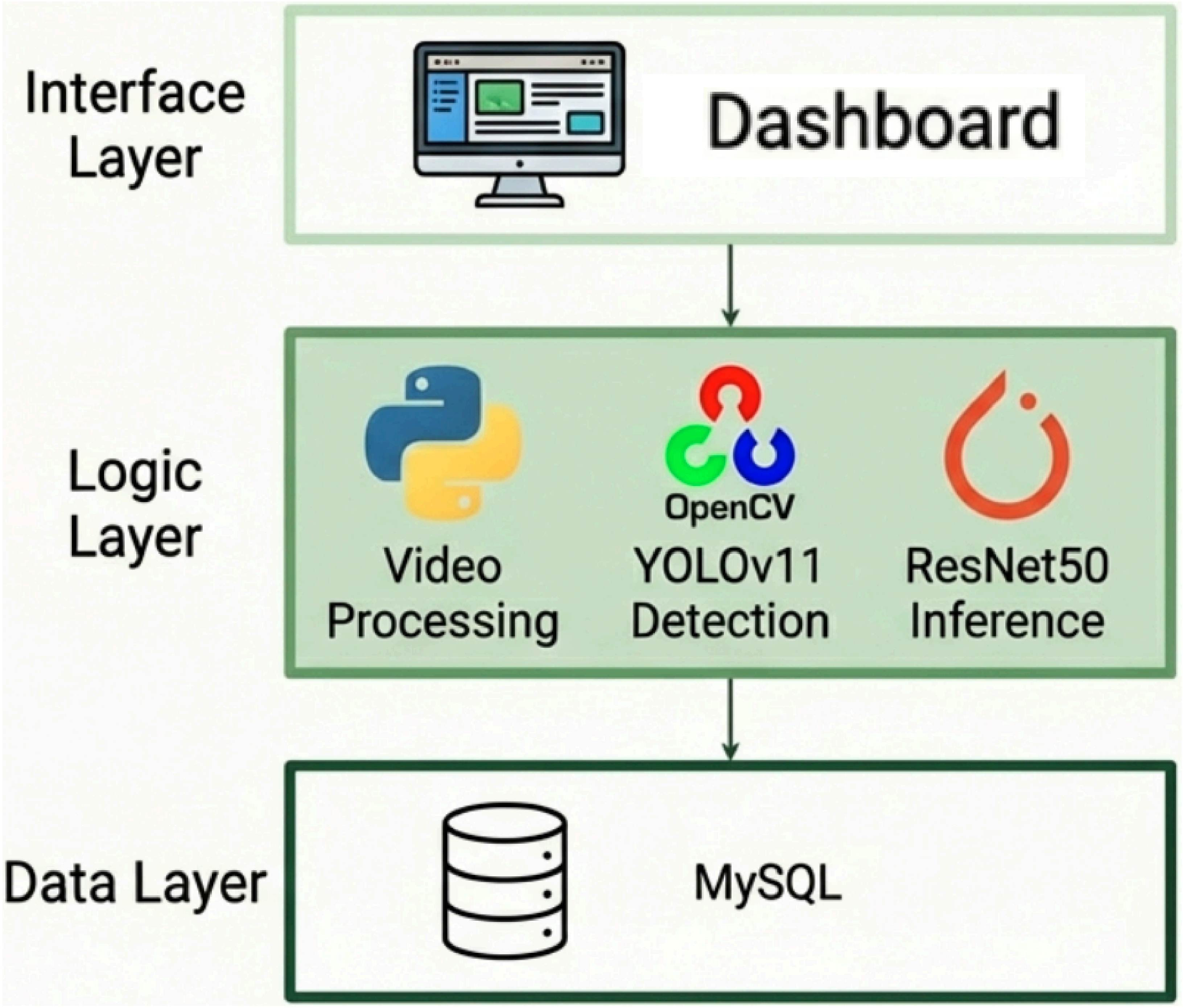
User interface and System Output

- Clean interface for rapid result inspection
- Flight summaries of detected diseased plants
- Plant-level diagnostics including disease type and severity
- Associates each detected plant with GPS coordinates for field reference





System Architecture Overview



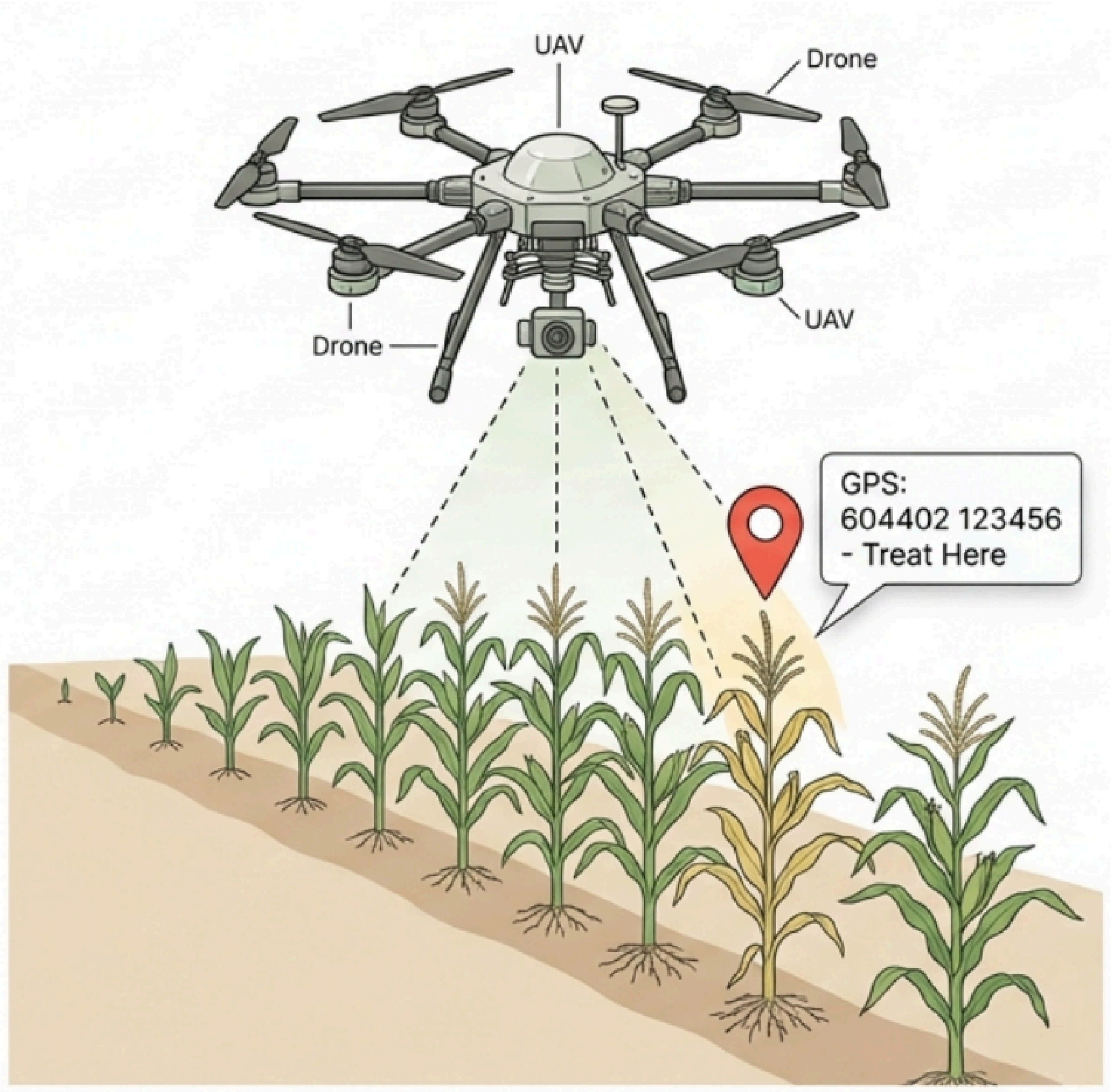
Evaluation and Verification KPIs

Key Performance Indicators	
Detection Accuracy	Compare bounding boxes with ground truth
Classification Accuracy	Evaluate disease classification using PlantVilalge and field data
Tracking Stability	Measure identity persistence and number of ID switches across frames
System Efficienty	Measure end-to-end processing speed
Field Coverage	Target: 98% detection rate for diseased plants

Expected Challenges:

- **Scale & Resolution:** Leaves occupy small regions within wide-angle UAV frames
- **Occlusion:** Dense foliage causes partial visibility and overlap
- **Identity Continuity:** Leaves may disappear and reappear due to drone motion
- **Environmental Noise:** Wind, lighting , and vibration degrade image quality

Innovation: Bridging Lab AI and Field Reality



Impact Benefits:

- **Reduced Chemical Usage:** Enables spot treatment instead of blanket spraying
- **Cost Efficient:** Lowers operational and input costs for farmers
- **Scalability**: Supports automated monitoring of large-scale agricultural fields